Predicting Cardiovascular Disease Using Machine Learning

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***Abstract*—Cardiovascular diseases remain among the leading causes of death worldwide, affecting millions annually and causing severe strain on healthcare systems. This project applies machine learning techniques to predict cardiovascular risk based on behavioral, demographic, and clinical factors. Through systematic preprocessing, feature selection, and model evaluation, this study demonstrates how data-driven approaches can support more proactive and personalized health risk assessment.**

I. Introduction

This work aims to predict the risk of Myocardial Infarction or Coronary Heart Disease (MICHD) using Behavioral Risk Factor Surveillance System (BRFSS) data [1]. The dataset covers diverse behavioral, demographic, and clinical features. Our goal is to identify the most effective modeling approach for cardiovascular risk prediction, emphasizing optimization of F1-score and accuracy for balanced performance.

II. Data Exploration

The dataset contains 328,135 samples and 321 features spanning behavioral, demographic, and clinical aspects. Over 47M entries are missing, and the target is strongly imbalanced (8.83% positive). This highlights the need for robust preprocessing and data balancing to ensure fair and reliable model training.

III. Data Preprocessing

A crucial step before training the predictive model consisted in transforming the raw survey data into a consistent and informative structure suitable for optimal machine learning.

Handling Missing and Invalid Entries A first essential step was to address missing and invalid entries. According to the dataset documentation, specific numerical values such as 7, 77, 777, 9, 99, or 999 were used to represent non-responses unknown answers or simply invalid entries. These placeholder values were uniformly replaced with NaN to allow for standardized handling.

Removing Features with Excessive Missing Values The plot below 1 motivates the removal of features with excessive missingness. To maintain dataset completeness while ensuring feature reliability, columns with more than 99% missing values were removed. This conservative threshold preserved a wide range of potentially informative features while eliminating those too incomplete to contribute meaningfully to model learning, thereby reducing noise and bias in the dataset.

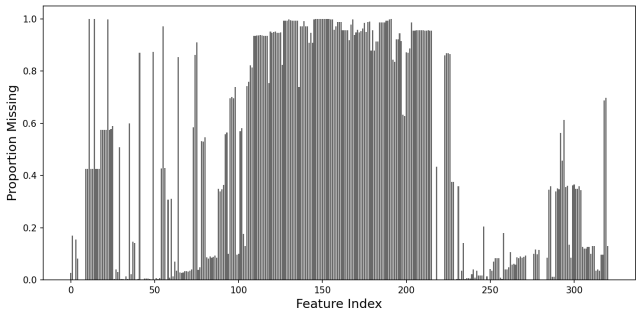


Figure 1: Proportion of missing values per feature

Identifying Numerical and Categorical Features Features were divided into numerical and categorical types based on their number of unique values: variables with ≤ 100 distinct entries were treated as categorical, while the rest were treated as numerical.

Missing Value Imputation For numerical features, missing values were imputed with the *median*, a robust estimator that mitigates the influence of extreme observations. For categorical variables, missing entries were imputed with the *mode*, preserving the most frequent and representative category.

Outlier Removal Extreme observations were trimmed by excluding the lowest 0.05% and highest 0.05% quantiles of each numerical feature. This allows to remove aberrant or unrealistic values.

Standardization All numerical features were standardized to zero mean and unit variance to prevent variables with large scales from dominating the optimization.

Low Variance and Highly Correlated Features Features with variance below 10^{-2} or correlation $|r| > 0.95$ were removed to reduce redundancy and improve model efficiency.

IV. Predictive Modeling and Evaluation

After preprocessing and exploring the dataset, the next step consisted in training the predictive classification models.

Evaluation Metrics Accuracy can be misleading under class imbalance (91% for the majority class). Thus, the **F1-score**, balancing precision and recall, was used as the main metric.

Logistic Regression

- **Motivation:** Logistic regression was naturally the first model adopted for this binary classification task. It served as a baseline due to its simplicity and interpretability.
- **Hyperparameter Tuning:** To optimize model performance, we systematically explored combinations of the learning rate γ and the regularization strength λ. The learning rate controls the convergence behavior of the gradient descent algorithm, where overly small values slow training and excessively large values can lead to divergence. The regularization parameter λ prevents overfitting by penalizing large coefficient magnitudes, thus improving generalization to unseen data.

Ridge Regression

- **Motivation:** Ridge regression was selected for its ability to handle high-dimensional data and to enhance model generalization through ℓ_2 regularization. By penalizing large coefficient magnitudes, it prevents overfitting and ensures that the model remains stable.
- **Parameter Optimization and Threshold Selection:** Model tuning involved jointly exploring λ, the degree of polynomial expansion, and the decision threshold. Since ridge regression outputs continuous scores, the threshold was optimized, rather than fixed at 0, to maximize the F1 score and achieve a balanced trade-off between precision and recall under class imbalance.

V. Results and Performance Evaluation

The results presented in the tables indicate that both ridge and logistic regression models delivered consistent performance across different hyperparameter settings. Ridge regression achieved a slightly higher F1-score, peaking at 0.4246 compared to 0.411 for logistic regression, while maintaining similar accuracy levels. Due to its higher F1-score and lower computational cost, ridge regression was selected for further tuning. After extended search and threshold optimization, it achieved 0.4278 on validation and 0.440 on the AIcrowd platform.

VI. Discussion and Analysis

When iterating on preprocessing thresholds, we observed that retaining most features while removing only those clearly unsuitable for model training led to the best overall performance. See Table II for ablations. Another key improvement was using a weighted loss in ridge regression to handle class imbalance. Unlike oversampling, which increases redundancy and risk of overfitting to duplicated minority samples, weighted loss directly increases the model's sensitivity to minority cases.

Evaluation on test data indicates that the model correctly identifies 82.3% of negative cases and 4.8% of positive cases. Misclassifications occur with 8.9% false positives and 4.0% false negatives. Given the medical context, false negatives (missing individuals at risk) pose a greater concern, as early detection of cardiovascular condition is critical. In practical setting, the decision threshold could be lowered even more to further reduce false negatives while accepting higher rate of false detections.

Table I: Comparison of Ridge and Logistic Regression performance across hyperparameters.

(a) Ridge regression (λ)

λ	Accuracy (test)	Accuracy (train)	F1 (test)	F1 (train)
10^{-6}	0.8703	0.8706	0.4231	0.4338
10^{-5}	0.8705	0.8708	0.4233	0.4338
10^{-4}	0.8703	0.8707	0.4224	0.4338
10^{-3}	0.8672	0.8678	0.4226	0.4335
10^{-2}	0.8721	0.8718	**0.4246**	0.4336
10^{-1}	0.8525	0.8519	0.4219	0.4334

(b) Logistic regression (λ, γ)

(λ, γ)	Accuracy (test)	Accuracy (train)	F1 (test)	F1 (train)
(0, 0.01)	0.742	0.742	0.211	0.219
(0.001, 0.01)	0.747	0.748	0.213	0.220
(0.1, 0.01)	0.861	0.862	**0.411**	0.416
(0, 0.1)	0.845	0.847	0.384	0.393
(0.001, 0.1)	0.857	0.859	0.396	0.406
(0.1, 0.1)	0.867	0.869	0.410	0.418
(0, 1.0)	0.819	0.822	0.315	0.326
(0.001, 1.0)	0.826	0.828	0.323	0.335
(0.1, 1.0)	0.864	0.866	0.398	0.407

Table II: Ablation study on preprocessing steps.

Preprocessing steps	**Mean F1-Score**	**Diff**
None	0.42636	0.0000 $\pm$0.0008
w/o replacing special values	0.42578	-0.0006 $\pm$0.0003
w/o polynomial expansion	0.42536	-0.0010 $\pm$0.0008
w/o num/cat feature distinction	0.42535	-0.0010 $\pm$0.0011
w/o LDA features	0.42513	-0.0012 $\pm$0.0007

VII. Conclusion

This project demonstrated how machine learning techniques can be effectively applied to predict cardiovascular disease risk using large-scale behavioral and clinical survey data. Through systematic preprocessing, feature selection, and model evaluation, results highlight the potential of data-driven methods for supporting early detection and preventive healthcare in cardiovascular risk assessment.

REFERENCES

[1] C. for Disease Control and P. (CDC), "Behavioral risk factor surveillance system: 2015 codebook report (land-line and cell-phone data)," U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, Atlanta, GA, Tech. Rep., August 2016, available online: https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf.

[1]